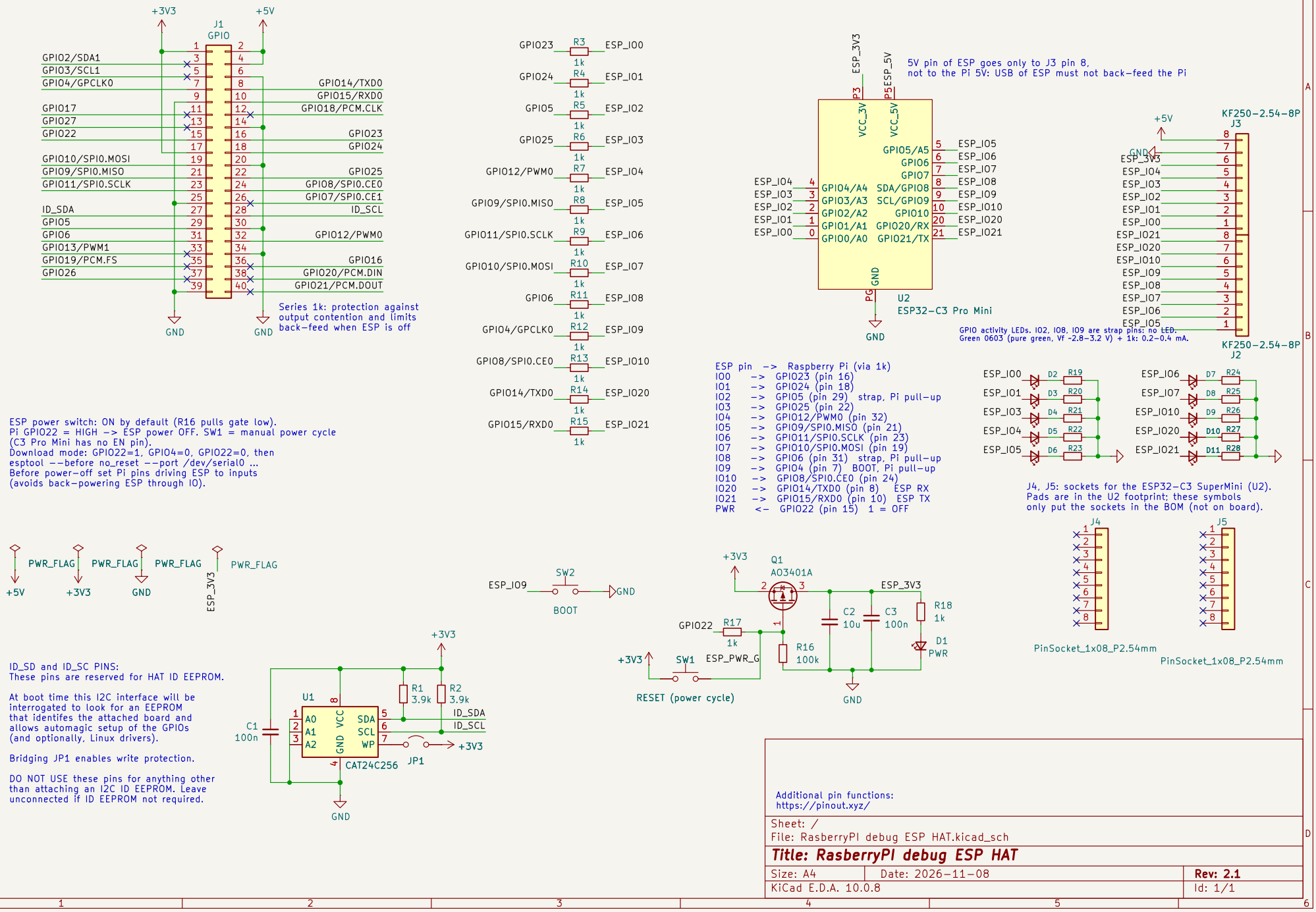

J1
GPIO
+3V3
+5V
GND
GPIO2/SDA1
GPIO3/SCL1
GPIO4/GPCLK0
GPIO17
GPIO27
GPIO22
GPIO10/SPI0.MOSI
GPIO9/SPI0.MISO
GPIO11/SPI0.SCLK
ID_SDA
GPIO5
GPIO6
GPIO13/PWM1
GPIO19/PCM.FS
GPIO26
GPIO14/TXD0
GPIO15/RXD0
GPIO18/PCM.CLK
GPIO23
GPIO24
GPIO25
GPIO8/SPI0.CE0
GPIO7/SPI0.CE1
ID_SCL
GPIO12/PWM0
GPIO16
GPIO20/PCM.DIN
GPIO21/PCM.DOUT
Series 1k: protection against output contention and limits back-feed when ESP is off
GPIO23 R3 1k ESP_IO0
GPIO24 R4 1k ESP_IO1
GPIO5 R5 1k ESP_IO2
GPIO25 R6 1k ESP_IO3
GPIO12/PWM0 R7 1k ESP_IO4
GPIO9/SPI0.MISO R8 1k ESP_IO5
GPIO11/SPI0.SCLK R9 1k ESP_IO6
GPIO10/SPI0.MOSI R10 1k ESP_IO7
GPIO6 R11 1k ESP_IO8
GPIO4/GPCLK0 R12 1k ESP_IO9
GPIO8/SPI0.CE0 R13 1k ESP_IO10
GPIO14/TXD0 R14 1k ESP_IO20
GPIO15/RXD0 R15 1k ESP_IO21
ESP_3V3
ESP_5V
P3
P5
5V pin of ESP goes only to J3 pin 8,
not to the Pi 5V: USB of ESP must not back-feed the Pi
VCC_3V
VCC_5V
GPIO5/A5
GPIO6
GPIO7
GPIO4/A4 SDA/GPIO8
GPIO3/A3 SCL/GPIO9
GPIO2/A2 GPIO10
GPIO1/A1 GPIO20/RX
GPIO0/A0 GPIO21/TX
ESP_IO4 4
ESP_IO3 3
ESP_IO2 2
ESP_IO1 1
ESP_IO0 0
5 ESP_IO5
6 ESP_IO6
7 ESP_IO7
8 ESP_IO8
9 ESP_IO9
10 ESP_IO10
20 ESP_IO20
21 ESP_IO21
P6
GND
U2
ESP32-C3 Pro Mini
GPIO activity LEDs. IO2, IO8, IO9 are strap pins: no LED.
Green 0603 (pure green, Vf ~2.8–3.2 V) + 1k: 0.2–0.4 mA.
KF250-2.54-8P
J3
+5V
GND
ESP_3V3
ESP_IO4
ESP_IO3
ESP_IO2
ESP_IO1
ESP_IO0
ESP_IO21
ESP_IO20
ESP_IO10
ESP_IO9
ESP_IO8
ESP_IO7
ESP_IO6
ESP_IO5
KF250-2.54-8P
J2
ESP pin -> Raspberry Pi (via 1k)
IO0 -> GPIO23 (pin 16)
IO1 -> GPIO24 (pin 18)
IO2 -> GPIO5 (pin 29) strap, Pi pull-up
IO3 -> GPIO25 (pin 22)
IO4 -> GPIO12/PWM0 (pin 32)
IO5 -> GPIO9/SPI0.MISO (pin 21)
IO6 -> GPIO11/SPI0.SCLK (pin 23)
IO7 -> GPIO10/SPI0.MOSI (pin 19)
IO8 -> GPIO6 (pin 31) strap, Pi pull-up
IO9 -> GPIO4 (pin 7) BOOT, Pi pull-up
IO10 -> GPIO8/SPI0.CE0 (pin 24)
IO20 -> GPIO14/TXD0 (pin 8) ESP RX
IO21 -> GPIO15/RXD0 (pin 10) ESP TX
PWR <- GPIO22 (pin 15) 1 = OFF
ESP_IO0 D2 R19
ESP_IO1 D3 R20
ESP_IO3 D4 R21
ESP_IO4 D5 R22
ESP_IO5 D6 R23
ESP_IO6 D7 R24
ESP_IO7 D8 R25
ESP_IO10 D9 R26
ESP_IO20 D10 R27
ESP_IO21 D11 R28
J4, J5: sockets for the ESP32-C3 SuperMini (U2).
Pads are in the U2 footprint; these symbols
only put the sockets in the BOM (not on board).
J4
J5
PinSocket_1x08_P2.54mm
PinSocket_1x08_P2.54mm
ESP power switch: ON by default (R16 pulls gate low).
Pi GPIO22 = HIGH -> ESP power OFF. SW1 = manual power cycle
(C3 Pro Mini has no EN pin).
Download mode: GPIO22=1, GPIO4=0, GPIO22=0, then
esptool --before no_reset --port /dev/serial0 ...
Before power-off set Pi pins driving ESP to inputs
(avoids back-powering ESP through IO).
PWR_FLAG
+5V
PWR_FLAG
+3V3
PWR_FLAG
GND
PWR_FLAG
ESP_3V3
ESP_IO9
SW2
BOOT
GND
+3V3
Q1
AO3401A
ESP_3V3
GPIO22
R17
1k
R16
100k
C2
10u
C3
100n
R18
1k
D1
PWR
+3V3
SW1
ESP_PWR_G
RESET (power cycle)
GND
ID_SD and ID_SC PINS:
These pins are reserved for HAT ID EEPROM.
At boot time this I2C interface will be
interrogated to look for an EEPROM
that identifes the attached board and
allows automagic setup of the GPIOs
(and optionally, Linux drivers).
Bridging JP1 enables write protection.
DO NOT USE these pins for anything other
than attaching an I2C ID EEPROM. Leave
unconnected if ID EEPROM not required.
+3V3
U1
A0
A1
A2
GND
VCC
SDA
SCL
WP
CAT24C256
R1
3.9k
R2
3.9k
ID_SDA
ID_SCL
C1
100n
JP1
+3V3
GND
Additional pin functions:
https://pinout.xyz/
Sheet: /
File: RasberryPI debug ESP HAT.kicad_sch
Title: RasberryPI debug ESP HAT
Size: A4
Date: 2026-11-08
Rev: 2.1
KiCad E.D.A. 10.0.8
Id: 1/1